

Background

❖ Problems of flexible-link manipulator control

- Rigid FK cannot accurately predict settled positions
- Vive Tracker data used as settled-position labels
- Supervised dataset using commands, encoders, and targets
- Position prediction and joint-space diagnosis using density-aware local KRR

❖ Prediction pipeline



Research Objective

❖ Collection of tracker-referenced position error dataset

- Measure settled end-effector position using Vive Tracker
- Align coordinates using initial-pose reference and rotation matrix
- Build dataset from commands, encoders, and targets
- Remove missing, tracker-error, and unsettled samples

❖ Development of a position prediction model for flexible manipulator

- Input features: command angles, encoder angles, and target positions
- Supervised model using Vive Tracker settled positions as labels
- Local joint-space prediction using density-aware local KRR

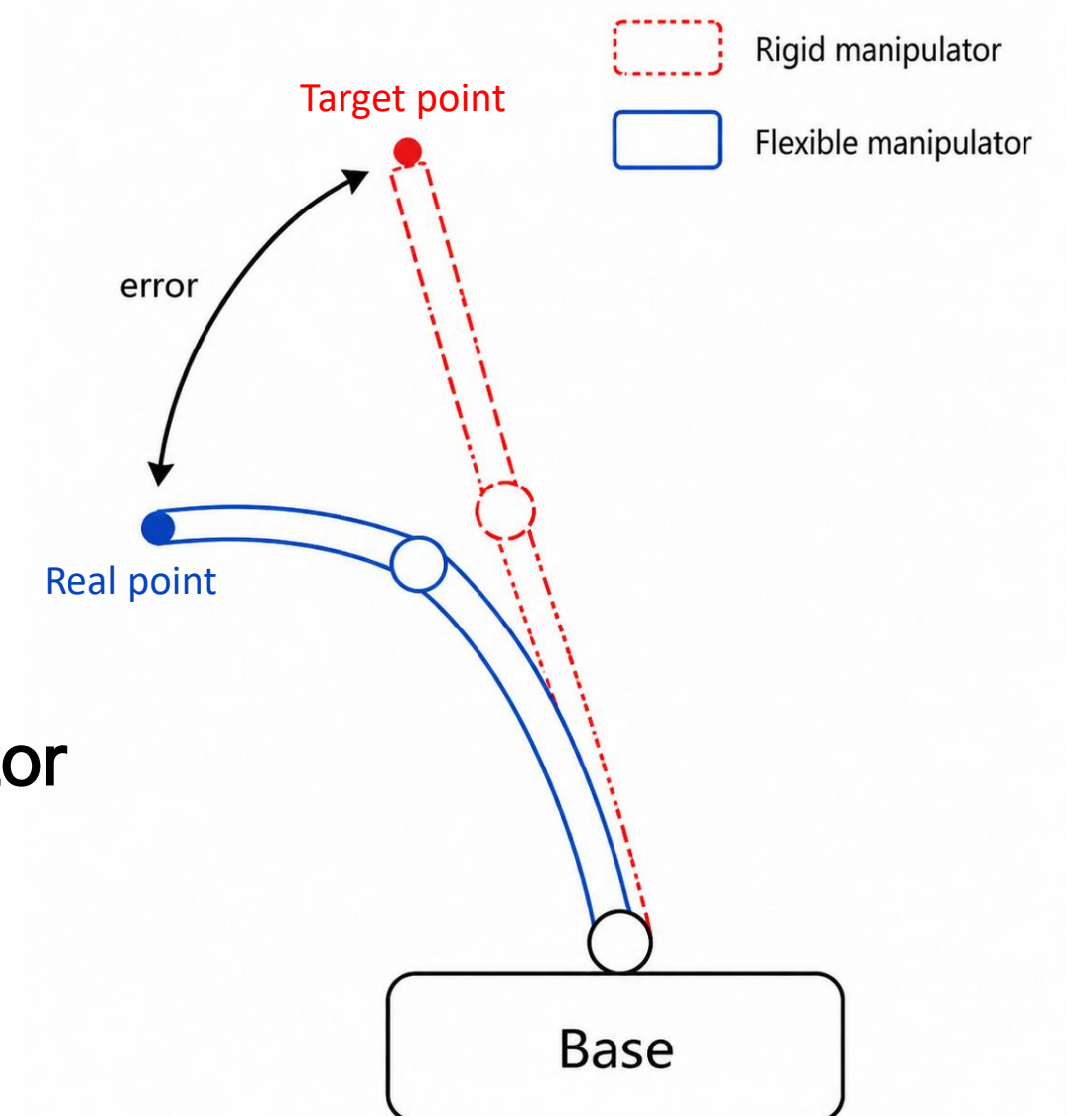


Fig 1. Error schematics of a flexible manipulator

Methodology

❖ Experimental setup

- SO-101 manipulator + rectangular TPU 95A link
- ST3215 servo + Waveshare ESP32 servo driver
- Dimensions of 1st flexible link : $40 \times 12 \times 300$ mm (W×H×L)
- Dimensions of 2nd flexible link : $40 \times 12 \times 180$ mm (W×H×L)
- HTC Vive Tracker 3.0 + Base station 2.0 × 2EA

❖ Data collection and preprocessing

- Measure settled end-effector position
- Synchronize commands, encoders, targets, and measurements
- Apply relative coordinates and frame alignment
- Valid samples: 584/600 for 1-link, 4,812/5,000 for 2-link



Fig 2. Rendering image of SO-101-based flexible TPU manipulator

Experimental data

❖ Target points vs. measured 3D end-effector points

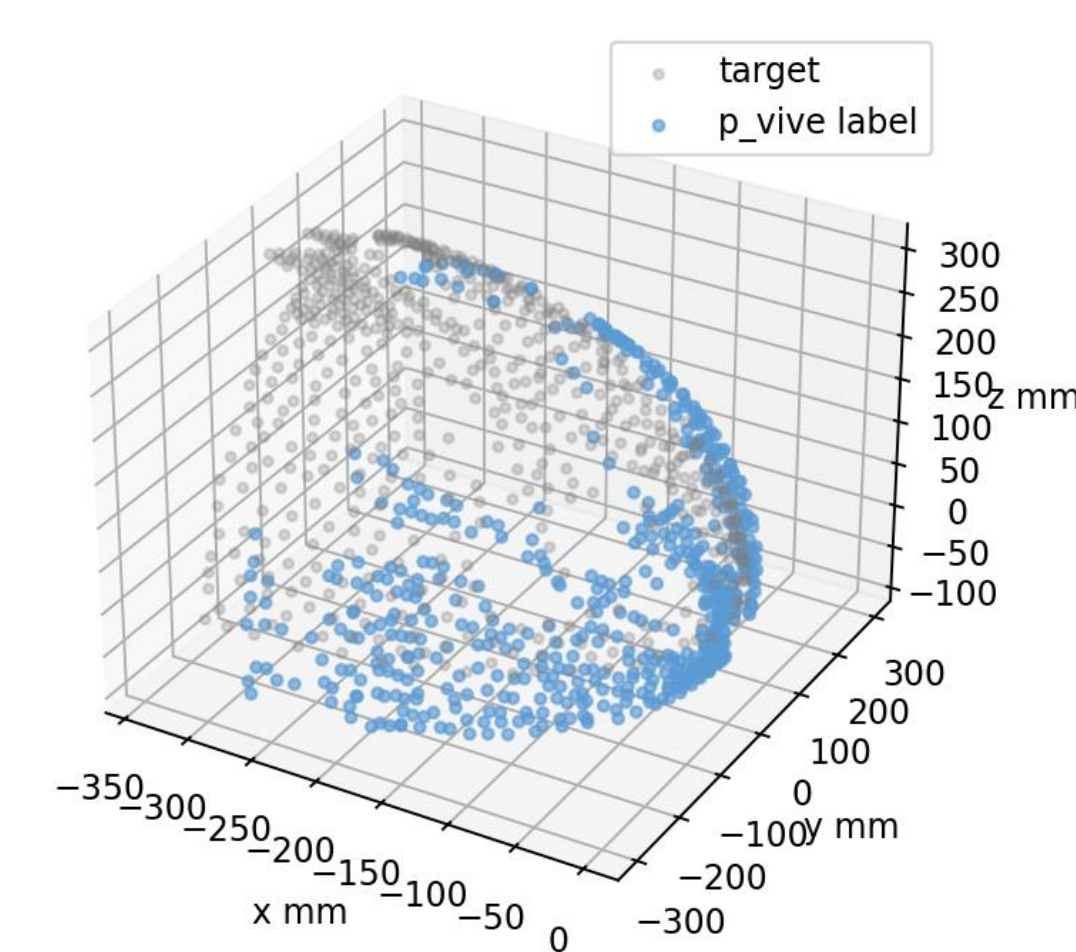


Fig 3. 1-link 3D point cloud dataset comparison

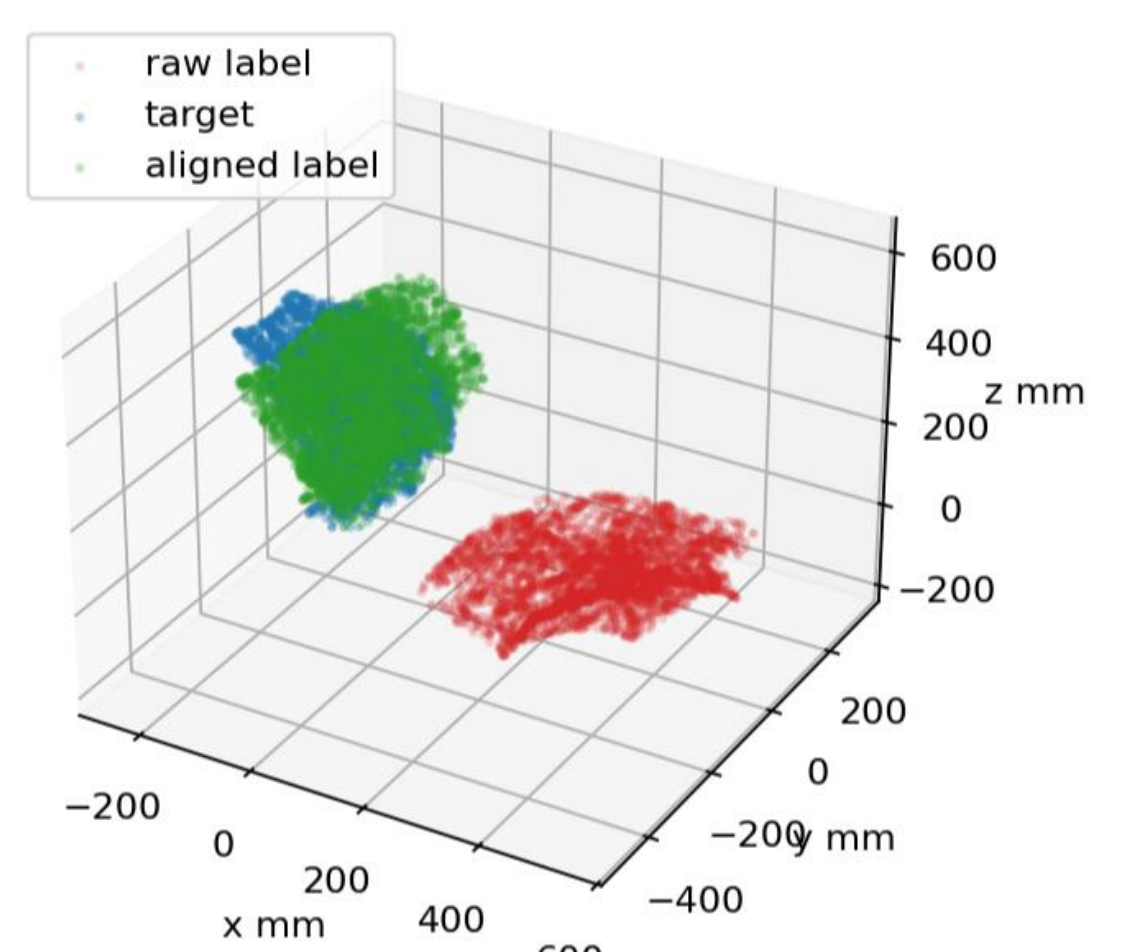


Fig 4. 2-link 3D point cloud dataset comparison

Results

❖ Coordinate alignment and preprocessing results

- Raw Vive Tracker coordinates showed axis-direction and origin mismatch with the target frame
- 1-link data were aligned by applying a **fixed rotation matrix** to the Tracker coordinate frame
- 2-link data were corrected using **Kabsch alignment** after data collection
- After Kabsch alignment, x/y/z-axis correlations **increased to 0.940 / 0.968 / 0.924**
- After alignment, x/y/z-axis RMSE decreased to 21.8 / 27.7 / 37.5 mm

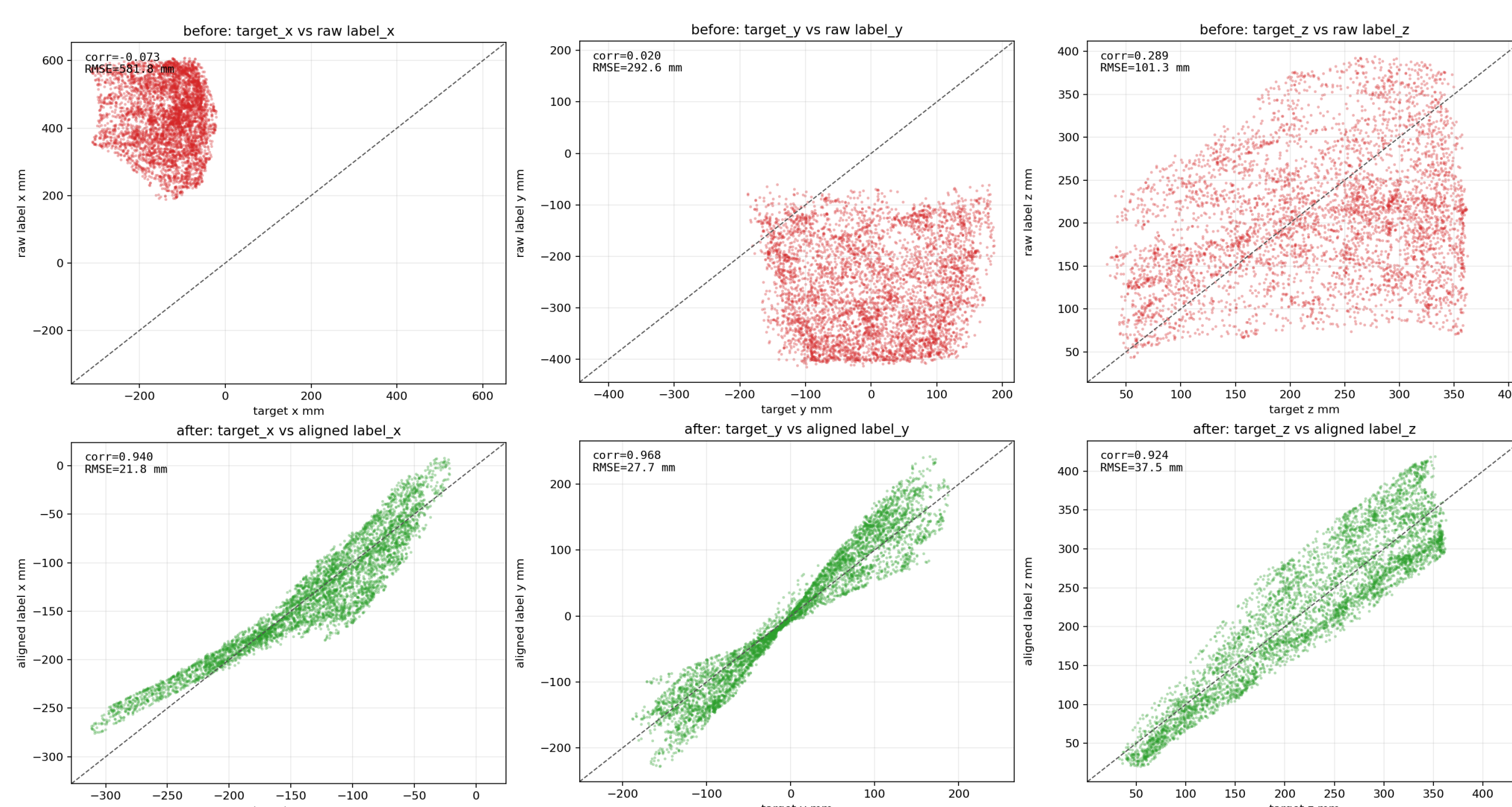


Fig 5. Comparison of 2-link raw dataset and Kabsch alignment dataset

❖ Position prediction model comparison

- Prediction performance was compared GPR baseline, RBF-KRR, and density-aware local KRR
- For the 1-link case, mean 3D prediction error **decreased from 5.97 mm to 2.01 mm**
- The 1-link model achieved a **median error of 1.57 mm** and pass@2 mm of 62.68%
- For the 2-link case, mean 3D prediction error **decreased from 18.82 mm to 0.71 mm**
- The 2-link model achieved a **median error of 0.52 mm** and pass@2 mm of 96.00%

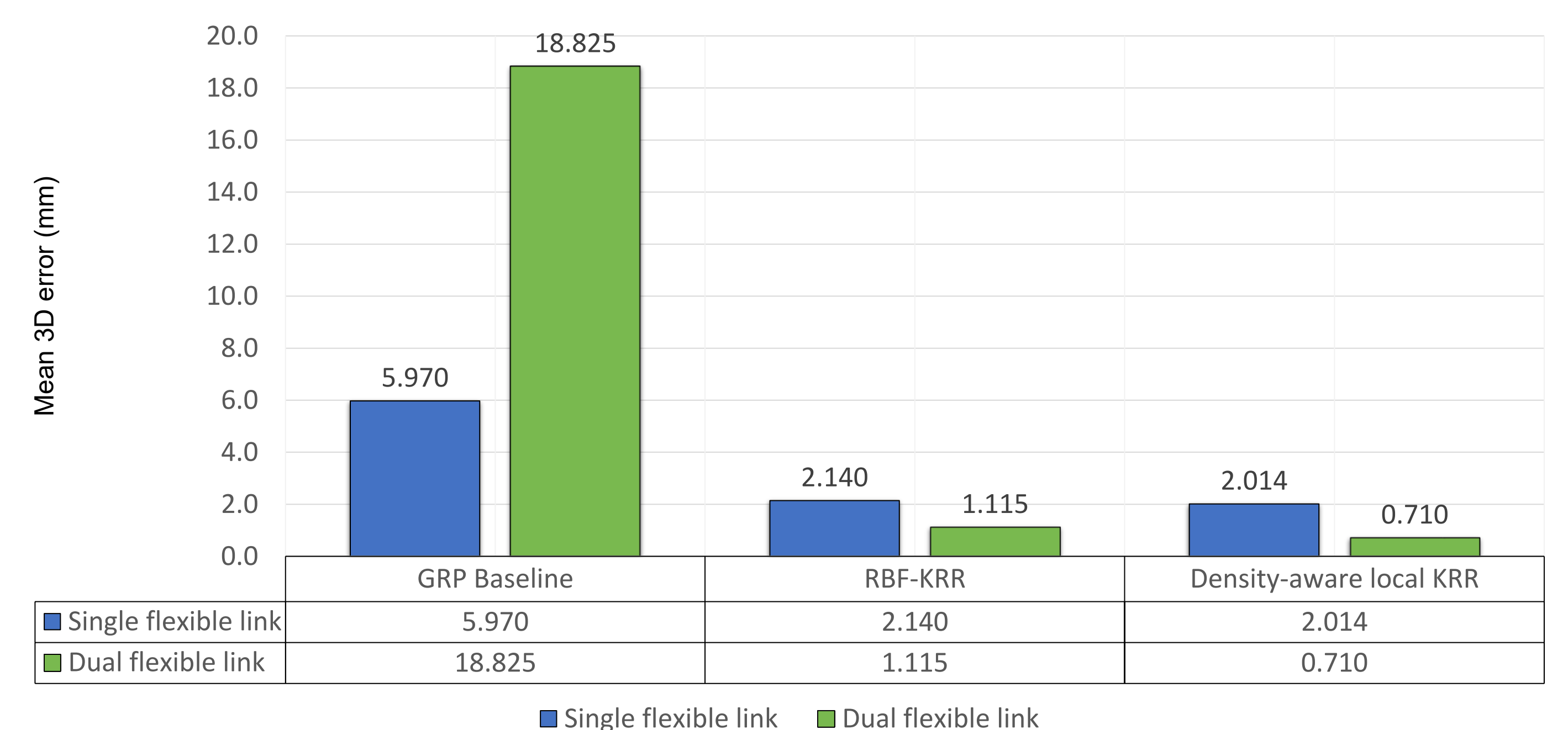
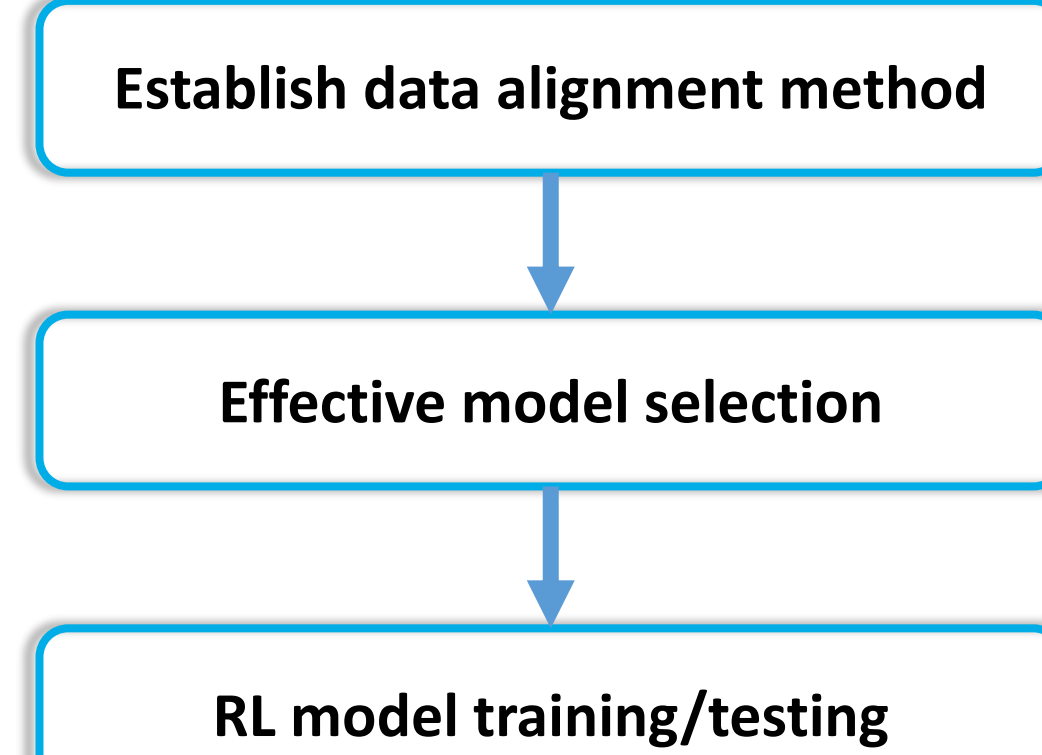


Fig 6. Mean 3D error comparison of Baseline/RBF-KRR/Density-aware local KRR prediction model

Discussion

❖ Effectiveness of Kabsch alignment and density-aware local KRR

- Established a data alignment method for flexible-link position prediction
- Built settled-position labels for learning-based compensation
- Verified Kabsch alignment for dual-link coordinate correction
- Density-aware local KRR achieved the lowest 3D prediction error
- Further hardware tests and joint-space diagnosis are needed for RL compensation



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